CHAPTER 3
RESEARCH METHODOLOGY

This chapter discusses the methodology used to develop **IronGrind: A Web-Based Fitness Gym Membership and Operations Management System for Iron Grind Fitness Gym in Purok Mangga, Sulongon, Dapitan City, Zamboanga del Norte**. It describes the development process, system requirements, design, data organization, and evaluation procedures. The system is intended to organize membership applications and records, membership periods and renewals, staff-recorded attendance, products, sales, and reports. Its development is organized using Agile, with work planned and delivered incrementally. Proposed evaluation activities are distinguished from results that have not yet been collected.

System Development Methodology

Agile is an iterative system-development approach in which stakeholder needs are gathered and prioritized, selected work is developed and tested in increments, and review results inform the next iteration. Figure 1 presents two connected flows for IronGrind. The circular cycle represents the repeated Develop, Test, and Review activities; feedback from a review informs the next increment. The lower project flow shows Data Gathering, Plan, and Design, followed by Launch only after the completed increments have been verified. The system work is organized around the gym's applicant/member and administrator workflows. The project does not assume a specific Agile framework such as Scrum; the research team should document the actual iteration length and review practices used and confirm that Agile is the methodology approved in the study proposal.

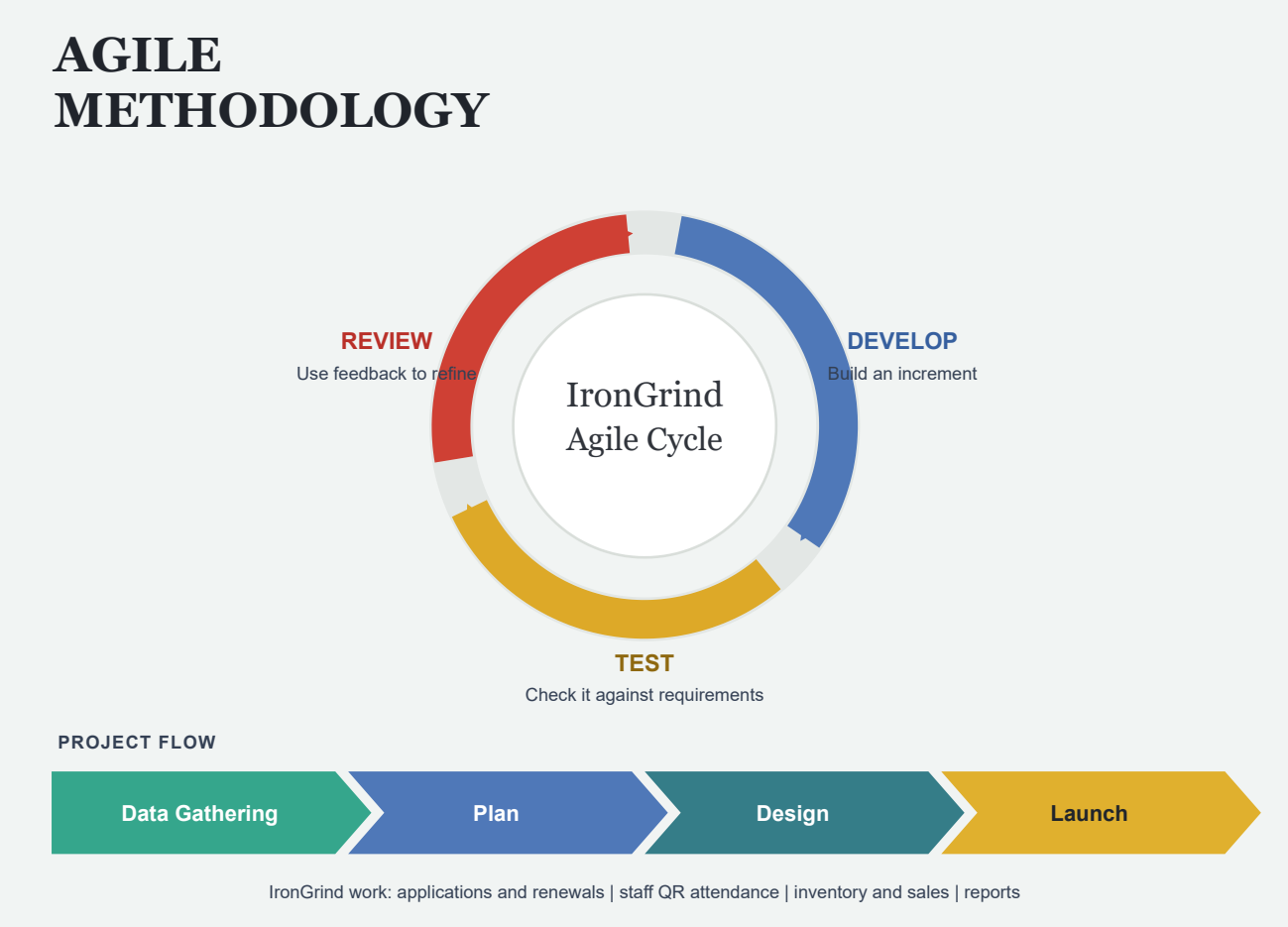


Figure 1. *Agile Methodology.*

Requirements Planning

Requirements planning begins with gathering information about the gym's workflows and needs from the owner and authorized staff and reviewing the project scope in Chapters 1 and 2. The information is translated into system requirements and organized in a prioritized task list, beginning with account access and membership applications, followed by staff review, member access, attendance, inventory, sales, and reporting. Each selected requirement is given acceptance criteria. The final study should identify the data-gathering techniques and participants actually used. Privacy and authorization requirements apply throughout development because the system handles personal and health-related information.

User Design

For each iteration, the relevant user workflows, page layouts, forms, reports, data relationships, and role permissions are designed or refined. The design distinguishes pending applications from approved member records, keeps attendance as a separate operational record, and stores sales with their itemized lines. The architecture, use-case, and data-flow figures in this chapter summarize the system design.

Prototypes

The system prototype is an interactive, browser-based version of the proposed gym workflows rather than a set of static screen mockups. Applicants can use the sign-in and registration pages to enter the application workflow. Members use the member portal to review membership information, display their QR code, view attendance history, and submit an eligible renewal request. Administrators use the dashboard to review applications and manage membership, staff-scanned attendance, products, sales, receipts, and reports. The prototype uses Supabase Auth and the configured database services for supported operations. Its key pages are listed in the User Interface Modules section. It should be treated as a prototype until deployment and operational readiness have been verified.

Construction

Selected requirements are implemented in the HTML, CSS, and JavaScript client and the Supabase authentication and PostgreSQL services. Database policies and functions support authorization and operational transactions. Each increment is tested against its acceptance criteria. At the end of an iteration, issues and feedback are recorded, priorities are updated, and unfinished or revised work returns to the prioritized task list. Stakeholder reviews or user feedback should be reported as completed only when the team has records of those activities.

Implementation

Launch takes place after the required increments pass verification; it is not repeated at the end of every iteration. The website and Supabase project can then be configured for the intended environment using the correct project settings, access policies, and authorized administrator accounts. Staff should verify the workflows and receive appropriate orientation. This chapter does not claim a production deployment, completed training, or user acceptance unless those activities are documented separately.

Requirement Specifications

Study Setting and Intended Users

The system is intended for Iron Grind Fitness Gym in Purok Mangga, Sulongon, Dapitan City, Zamboanga del Norte. Its intended users are applicants, members, authorized administrators, and walk-in customers served by staff. The database defines `admin` and `member` profile roles; it does not define a separate staff role.

System Requirements

The following requirements describe the website's supported functions and operating conditions. Email notifications are identified as planned work and are not treated as an available output.

Table 1. Functional Requirements

ID	Requirement	Description
FR-01	Authentication and access	Users authenticate through Supabase Auth. Administrator and member access is governed by profile roles and database authorization policies.
FR-02	Online application	An applicant submits registration details through the website. The application is stored for administrator review and is not automatically an active membership. If the applicant attempts to log in while the application is pending, the website displays the pending status.
FR-03	Application review	Authorized staff view, approve, or reject registration and renewal applications. The decision and application status are retained.
FR-04	Membership management	Authorized staff manage member records, plan and membership dates, status, payment information, and membership lifecycle records.
FR-05	Member portal	An approved member views membership status and progress, personal attendance history, and a QR code; updates are limited to supported account and personal-information fields.
FR-06	Renewal request	An eligible member can submit a renewal request for staff review. A request is not an automatic renewal. Chapter 2 specifies the five-calendar-day renewal window before and including the current expiry date.
FR-07	Attendance	Staff scan a member's QR code to record check-in and check-out. Members can view their own attendance history; the member does not record their own attendance.
FR-08	Product and inventory management	Authorized staff maintain products and inventory. Supplements use gram-based quantities; merchandise uses item quantities and stock tracking.
FR-09	Sales and receipts	Staff record in-person product and membership sales for members or walk-in customers. The system stores a sale record and its line items and supports receipt/report views.
FR-10	Operational reports	Authorized staff view and print or save the available membership, member-status, archive, payment, attendance, renewal, inventory, and sales reports. Report contents depend on the records entered into the system.

Table 2. Nonfunctional Requirements

Area	Requirement
Platform	The application is accessed through a modern browser with JavaScript enabled. The user interface is implemented as web pages and client-side scripts.
Connectivity	Authenticated database-backed functions require an internet connection to reach Supabase.
QR scanning	The staff device requires a camera and browser permission to scan QR codes. Production camera access should be served from a secure HTTPS origin.
Access control	Database access is restricted through authenticated roles and PostgreSQL Row Level Security policies. Privileged account creation is routed through the administrator-verified Edge Function. The deployed configuration must be tested before use.
Data handling	The application stores contact, emergency-contact, and health-related information. Access and retention practices should follow the gym's approved privacy procedures and applicable law. This description is not a certification of legal compliance.
Availability	Users depend on the availability of the website host, internet connection, and Supabase services. Operational backup and recovery procedures must be defined for the deployed environment.

Table 3. Software Requirements

Software	Requirement or purpose
Web browser	Current browser with JavaScript enabled for applicant, member, and administrator workflows.
Frontend	HTML, CSS, and JavaScript files served by a compatible web host.
Backend services	Supabase Auth and PostgreSQL, with the project schema, Row Level Security policies, and required database functions configured.
Edge Function	Supabase Edge Function runtime for administrator-authorized member account creation.
Development environment	Node.js 20 or later to run the repository's local static-file server. This local server is for development, not the production database backend.
QR scanning	Browser camera permission and a secure HTTPS origin for production QR scanning.

Table 4. Hardware Requirements

User or component	Requirement
Administrator or staff	Desktop computer, laptop, or compatible mobile device with a current browser and internet access. A camera-enabled device is needed when staff scan member QR codes.
Applicant or member	Smartphone, tablet, laptop, or desktop computer with a current browser and internet access.
Web hosting	Hosting capable of serving the static HTML, CSS, JavaScript, and image assets over HTTPS. Supabase provides the separate authentication and database services.

Table 5. System Inputs and Outputs

Type	Information	Description
Input	Account credentials	Email and password are submitted to Supabase Auth.
Input	Registration or renewal application	Applicant and plan details, contact and emergency-contact information, goals, health declarations, and signature where collected by the form.
Input	Attendance action	Staff scans a member QR code and records check-in or check-out.
Input	Product and inventory update	Administrator enters product details, price, unit, and merchandise stock quantity where applicable.
Input	Sale transaction	Staff enters customer type and name, products or membership, quantities, and transaction details.
Input	Report request	Administrator selects a report type and available filters or date range.
Output	Authentication and login result	Role-appropriate routing or an authentication error; pending applicants see their application status during login.
Output	Application and membership result	The recorded application status and, after staff review, the decision and updated membership information.
Output	Member portal information	Membership details, QR code, personal attendance history, and active product catalog permitted for the member.
Output	Attendance confirmation	Confirmation of staff-recorded check-in/check-out and permitted attendance history.
Output	Receipt	Sale details, line items, quantities, prices, customer information, and total.

Type	Information	Description
Output	Administrative reports	Membership, member status, archives, payments, attendance, renewals, inventory, and sales data based on stored records.
Planned output	Email notifications	Application or renewal decisions and expiry reminders are planned; email delivery is not currently available.

System Design

Data Process Modeling

Data process modeling describes how information enters IronGrind, which processes handle it, where records are stored, and what results are returned to applicants, members, and administrators. The model is based on the current Supabase schema and implemented workflows. Email notification is shown only as a planned feature.

Diagrams

System Architecture

Figure 2 shows applicants and members using the login, registration, and member pages, while administrators use the admin dashboard. Supabase Auth verifies credentials and creates the authenticated session; the application then routes users to the member or administrator experience based on profile role. The browser modules are organized as Applications and Membership, Attendance and QR, Inventory and Products, Walk-in POS and Receipts, Member Portal, and Reports and Exports. These modules send requests through the Supabase Data API or database RPC functions. PostgreSQL Row Level Security (RLS) and function-level checks govern access to the Supabase PostgreSQL database, which stores profiles, membership applications, members, attendance, products, sales, sale items, and membership history. The current Realtime configuration publishes changes from `membership_applications` and `attendance` to subscribed website views. The diagram also shows email notification delivery as in development, with no external email provider selected; it should not be described as an operational feature until it is implemented and configured. The Node.js server in the repository serves static files for local development and is not the production database backend.

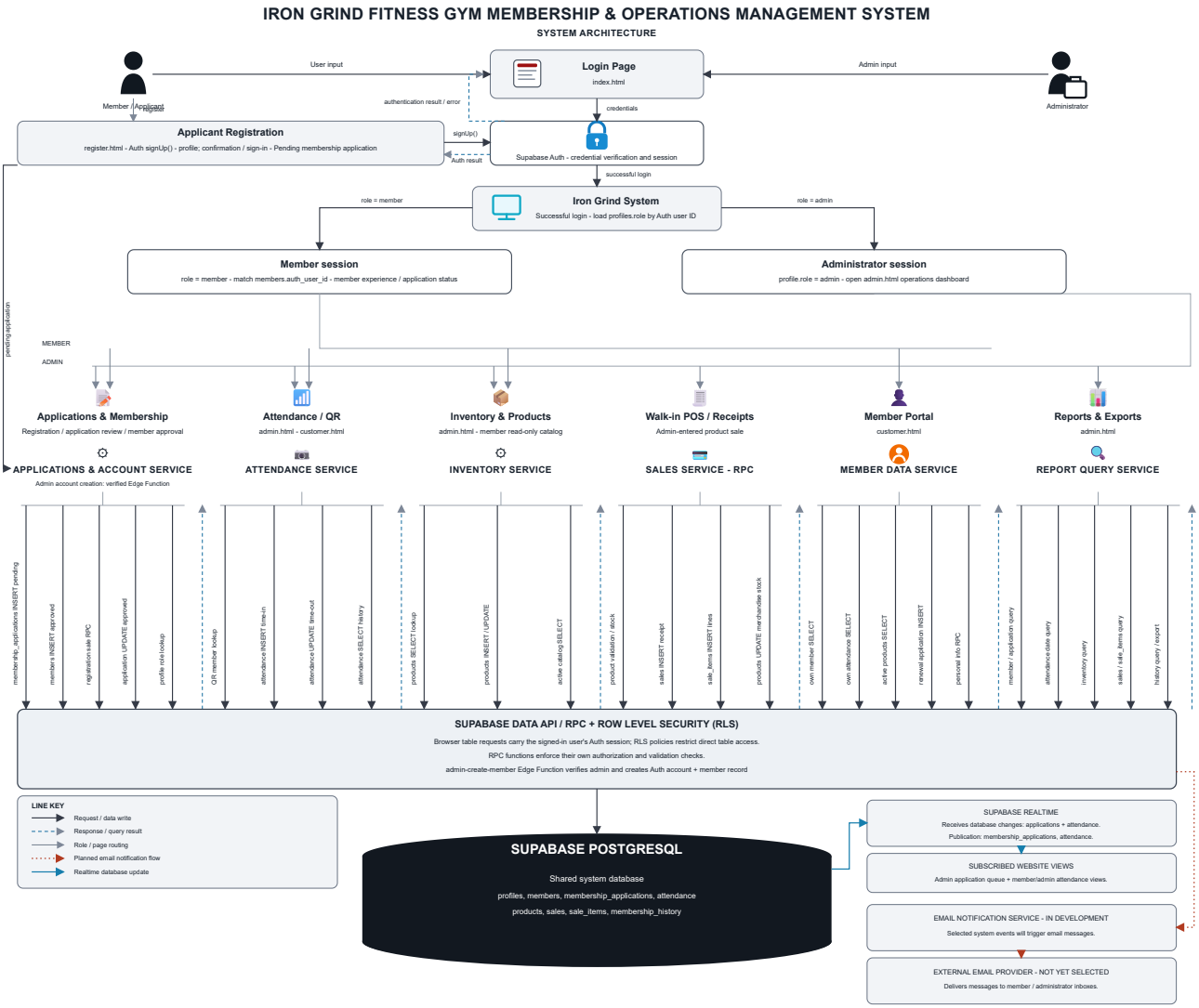


Figure 3. System Architecture Diagram

Figure 2. *System architecture of IronGrind.*

Context Diagram

The context diagram treats IronGrind as one system and shows the information exchanged with external actors and backend services. Applicants and members submit credentials, registration or renewal details, and supported profile requests; they receive authentication results and permitted membership, attendance, and catalog information. Administrators submit review decisions, operational updates, sale entries, and report filters; they receive records, receipts, and reports. Walk-in purchase details are entered by staff. Supabase Auth and Supabase data services provide authentication and authorized data access. Email notification integration is planned, not operational.

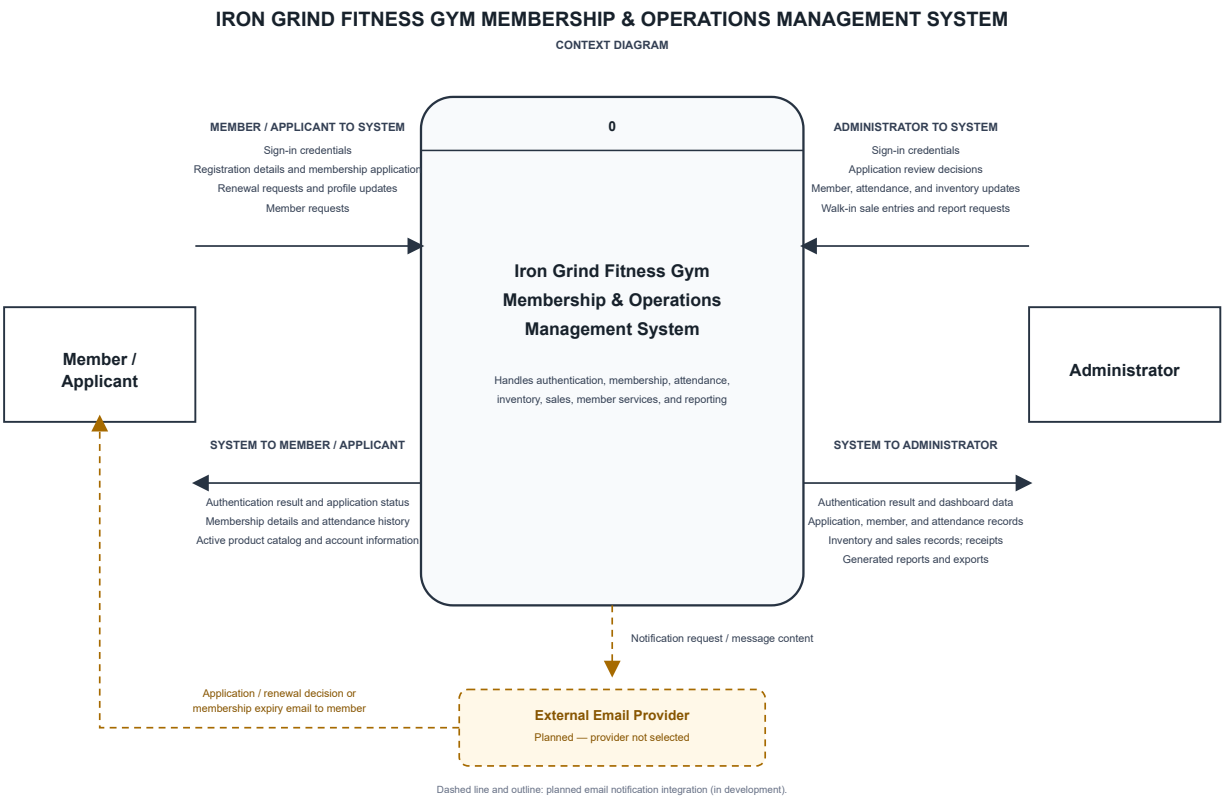


Figure 3. *System context diagram of IronGrind.*

Data Flow Diagram

The Level 1 data-process model shows how information moves between the member/applicant, administrator, system processes, and Supabase data stores. It covers authentication, membership applications, the member portal, staff-recorded attendance, inventory, walk-in sales, and reports. Email notifications are included as planned only; no provider has been selected. Solid arrows show requests or data writes, and dashed arrows show returned responses.

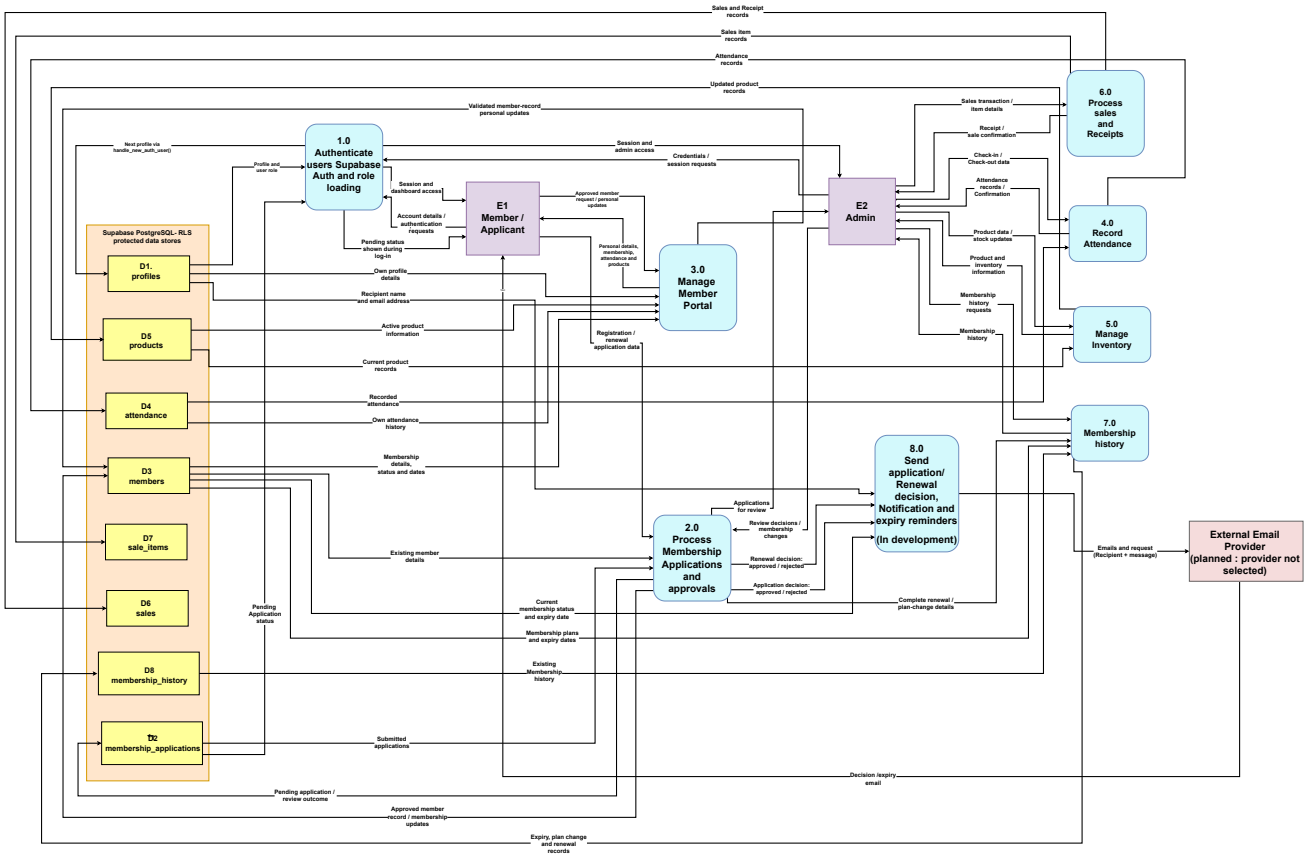


Figure 4. *Level 1 data-process model for IronGrind.*

Use-Case Diagram

The use-case diagram presents applicant/member and administrator interactions with IronGrind. Applicants and members use the account, application, portal, and attendance-history functions available to their roles. Administrators review applications and manage membership, attendance, products, sales, and reports. Walk-in transactions are entered by staff.

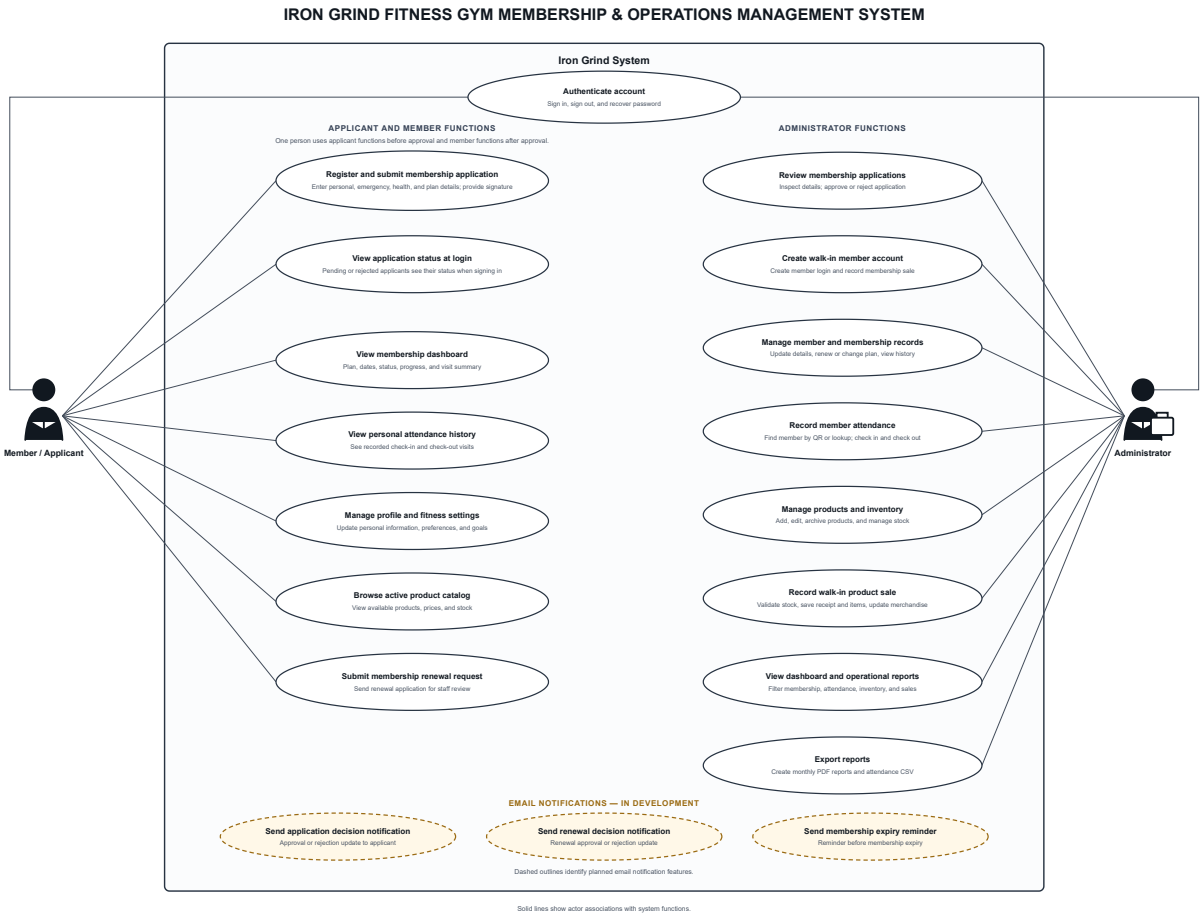


Figure 5. *Use-case diagram of IronGrind.*

Entity Relationship Diagram

The entity relationship diagram (ERD) summarizes the principal relationships in the Supabase PostgreSQL schema. A member can have multiple linked applications, attendance events, membership-history records, and sales. Each sale can have multiple sale items; a sale item may reference a product. Supabase Auth manages account identities separately. The member-to-Auth link is optional and unique, so an Auth user can be linked to at most one member.

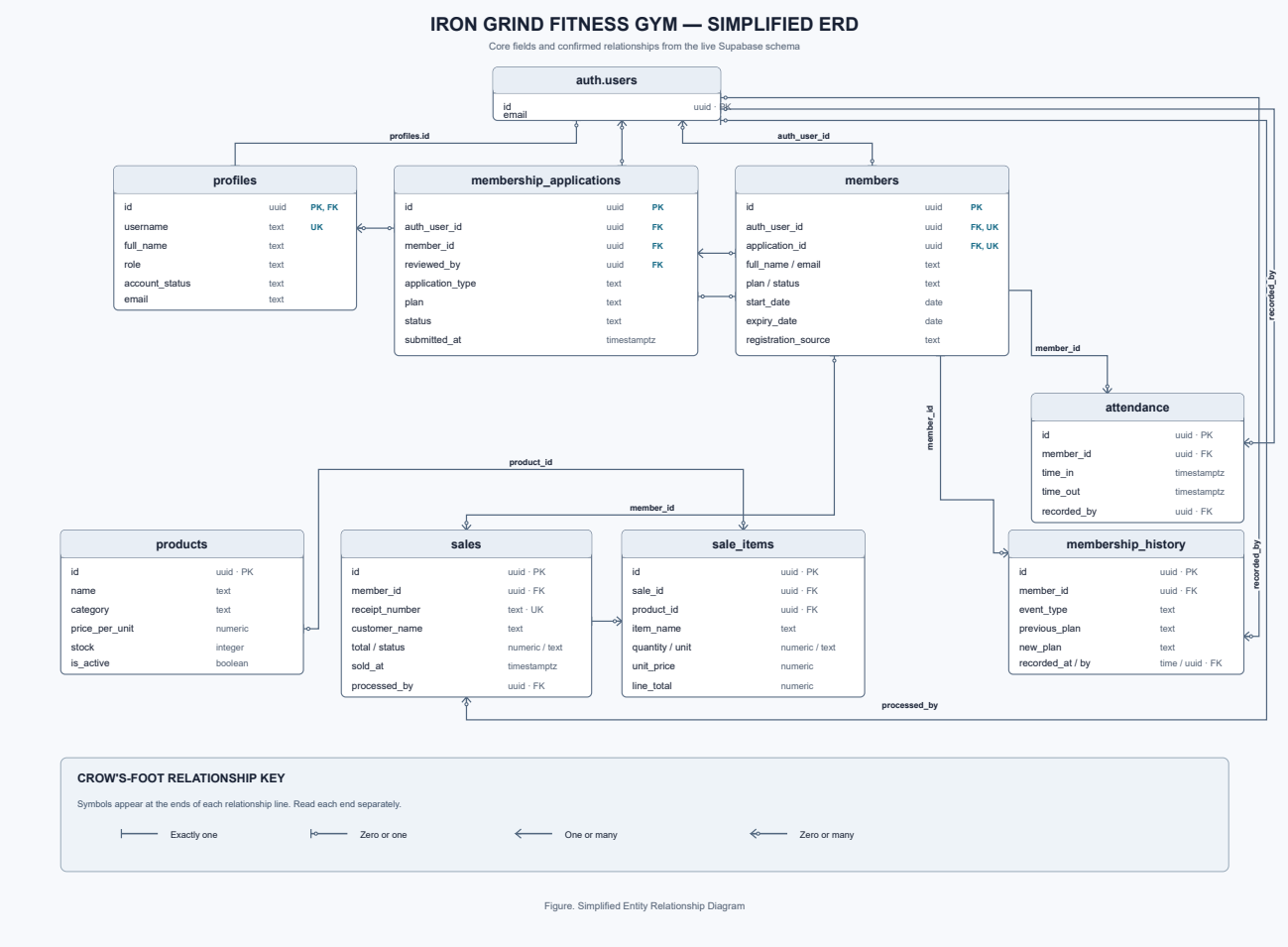


Figure 6. *Entity relationship diagram of IronGrind.*

Data Processes

Process	Input	Data stores or services used	Output
1.0 Authenticate User	Member/applicant or administrator credentials	Supabase Auth; D1 `profiles` for role lookup	Authenticated session and role-based routing, or an authentication error.
2.0 Process Membership Applications	Registration or renewal details; administrator approval or rejection	D2 `membership_applications`; D3 `members`; D8 `membership_history`; D6 `sales` when a membership transaction is recorded	Updated application status, member record and related transaction result.
3.0 Manage Member Portal	Member requests, supported profile updates, renewal submissions	D1 `profiles`; D2 `membership_applications`; D3 `members`; D4 `attendance`; D5 `products`	Member's permitted membership, attendance, and active product information; renewal or update result.
4.0 Record Attendance	Staff QR scan and check-in/check-out action	D3 `members` for member lookup; D4 `attendance` for attendance records	Saved attendance event and confirmation; attendance history for permitted member access.
5.0 Manage Inventory	Administrator product, price, category, or stock changes	D5 `products`	Updated product and inventory records; active catalog data for members.
6.0 Process Walk-in Sale	Staff-entered customer and sale-item details	D5 `products` for validation and stock; D6 `sales`; D7 `sale_items`	Validated sale, receipt details, and applicable merchandise stock update.

Process	Input	Data stores or services used	Output
7.0 Query Reports and History	Administrator report type and filters, including date range	D2 `membership_applications`; D3 `members`; D4 `attendance`; D5 `products`; D6 `sales`; D7 `sale_items`; D8 `membership_history`	Dashboard summaries, filtered report data, and available exports.
8.0 Email Notifications (planned)	Future approval or expiry events	No live notification store or external provider configured	No email output is currently produced; this process remains planned.

Data Stores

ID	Data store	Contents
D1	`profiles`	Authenticated user's application profile and role.
D2	`membership_applications`	Registration and renewal submissions, review status, and decision information.
D3	`members`	Approved member and membership records.
D4	`attendance`	Staff-recorded member check-in and check-out events.
D5	`products`	Product catalog, categories, prices, units, and merchandise stock.
D6	`sales`	Transaction headers and receipt details, including membership sales where recorded.
D7	`sale_items`	Itemized transaction lines.
D8	`membership_history`	Expiration, renewal, and plan-change events.